Infobits

Heating process in canning was first used by **Nicholas Appert** in 1890. He described a safe means of preserving all kinds of food substances in containers or in cans. Appert is known as father of cannning.

Heat resistance varies among different microorganisms. These differences can be expressed in terms of thermal death point. **Thermal Death Point (TDP)** is the lowest temperature at which all the microorganisms in a particular liquid suspension will be killed in 10 minutes.

Another factor to be considered in sterilization is the duration of time required. This is expressed as **Thermal Death Time (TDT)**. TDT is the minimal time required for all microorganism in a particular liquid culture to be killed at a given temperature. Both TDP and TDT are useful guidelines that indicate the degree of treatment required to kill a given population of bacteria.

Decimal Reduction Time (DRT) is related to bacterial heat resistance. DRT is the time, in minutes, in which 90% of a population of microorganism at a given temperature will be killed.

Heat is employed either as dry heat or moist heat.

4.4.1 Sterilization by Dry Heat

Dry heat is frequently used for the sterilization of glassware and laboratory equipments. In dry heat sterilization, microbial cells are apparently killed by oxidation of their constituents and protein denaturation. Dry heat is applied in the following ways:

a) Red heat

Inoculating wires, points of forceps and searing spatulas are sterilized by holding them in the flame of a bunsen burner until they are seen to be red hot.

b) Flaming

This method is used for sterilizing scalpels, needles, mouths of culture tubes, slides and cover slips. It involves passing the article through the bunsen flame without allowing it to become red hot.

c) Incineration

This is an excellent method for destroying materials such as contaminated clothes, cotton wool stoppers, animal carcasses and pathological materials. It involves burning of materials in incinerators.

d) Hot air oven

This is the most widely used method of sterilization using dry heat. The oven is usually heated by electricity and it has a thermostat that maintains the chamber air constantly at the chosen temperature.

It has a fan or turbo-blower to assist the circulation of air and to ensure rapid, uniform heating of the load. In Hot Air Oven, the air is heated at a temperature of 160°C for one hour. Figure 4.1 shows laboratory hot air oven.

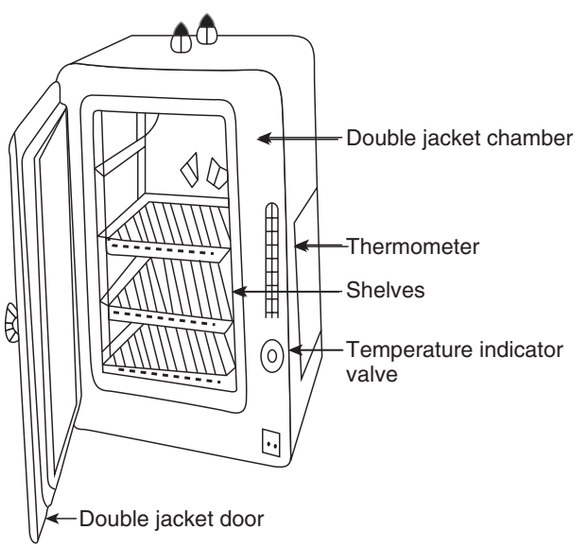


Figure 4.1: Hot Air Oven

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